

Xiaozhen Ma

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EDUCATION

Southwest Jiaotong University and University of Leeds

Sep 2023 - Jul 2027

Bachelor of Engineering in **Electronic and Electrical Engineering** (Joint Program, dual degree from both universities)

ADDITIONAL EDUCATION AND ACADEMIC PROGRAM

University of California, Irvine

Jun 2026 - Aug 2026

UCInspire Summer Research Programme

St Catharine's College, Cambridge

Feb 2025

Interdisciplinary Winter School (Computer Science, Data Science, and Artificial Intelligence Track)

RESEARCH INTERESTS

Computer Vision, 3D Reconstruction, Robotic Perception, Autonomous Driving

RESEARCH EXPERIENCES

PIKAN-Drive: Progressive Physics-Informed KANs from PilotNet Steering Regression to Fast-BEV 3D Detection

Project Lead, Supervised by Prof. Fadi Kurdahi, UC Irvine

Jun 2026 - Present

- Led a progressive study of MLP, KAN, and PI-KAN architectures from PilotNet steering regression to Fast-BEV multi-camera 3D detection, establishing unified training, evaluation, and ablation pipelines.
- Built a PilotNet benchmark on 45,406 sequential Udacity samples; the best PI-KAN achieved MAE 3.209 / RMSE 6.163, reducing error by 11.3% / 9.7% relative to the MLP baseline, while the Fast-BEV KAN baseline improved mAP from 0.2559 to 0.2720 and NDS from 0.3950 to 0.4046.
- Designed nine grouped physics-informed auxiliary losses and conducted targeted ablations for Fast-BEV; showed that additional constraints did not consistently improve detection, motivating task-specific, metric-aligned, and architecture-aware supervision.

Intelligent CAD Generation and Point-Cloud-to-Editable-CAD Reconstruction

Research Assistant, Supervised by Prof. Dong-Ming Yan and Prof. Mingyang Zhao, National Laboratory of Pattern

Recognition, Institute of Automation, Chinese Academy of Sciences

Feb 2026 - Present

- Deployed and evaluated CAD-Assistant for tool-augmented CAD generation through the FreeCAD Python API, analysing task planning, code generation, geometric modelling, and execution feedback; optimised prompts with dimensional and geometric constraints to improve code executability, solid validity, and parametric modelling quality.
- Reproduced Point2CAD and ParSeNet, implemented data-interface compatibility between the two pipelines, and investigated multi-view and three-axis projection boundary constraints, showing that additional boundary supervision did not reliably improve the official segmentation baseline through comparative experiments.
- Developed a CAD-Recode-based point-cloud-to-parametric-CAD pipeline, generating executable CadQuery programs from point-cloud inputs, executing and validating them through CadQuery and OpenCascade/OCF, and producing editable CAD programs together with B-Rep / STEP / Solid outputs.
- Designed and evaluated CAD-knowledge supervision, geometric verification, parameter refinement, structural diagnosis, and multi-candidate selection strategies; the independently confirmed S2@100 inference pipeline reduced median Chamfer Distance by 13.4% relative to S0@10 while also improving mean, P90, and mean IoU without reducing solid validity, establishing geometric candidate reranking as the strongest confirmed approach.

Sparse Point Cloud 3D Railway Scene Reconstruction Technology Driven by Monocular Depth Estimation

Project Lead, Supervised by Prof. Zhao Tang, State Key Laboratory of Rail Transit Vehicle System

Apr 2025 - May 2026

- Developed the FrustumDense-SAM-XC3D railway-scene 3D reconstruction system by integrating multi-frame LiDAR point clouds, RGB images, and monocular depth information into a pipeline covering point-cloud densification, depth fusion, object segmentation, occlusion handling, and scene visualization.
- Proposed a multi-frame frustum back-projection densification method that projected LiDAR points from the current, historical, and future frames into the camera frustum for spatial aggregation, filling sparse regions around buildings, tracks, and trackside facilities to improve point-cloud coverage and structural completeness in large-scale scenes.
- Integrated DepthPro-based monocular metric depth estimation with LiDAR pose constraints and applied patch-wise fitting, boundary resampling, and moving least-squares prediction to correct scale errors in distant regions of the monocular depth maps and generate pixel-level dense 3D point clouds.
- Applied the pretrained SAM3 model to multi-frame image segmentation and object-boundary detection, used a tree-based data structure to manage nested objects and occlusion relationships, and implemented cross-frame co-visibility consistency checks for point-cloud matching, clustering, and unique object identification.
- Designed class-aware radius filtering and patch-based rendering algorithms to suppress accumulated noise from multi-frame point-cloud fusion, extract tracks, buildings, and transportation facilities by category, and visualize large-scale railway scenes for single-frame object extraction and railway asset presentation.

Sparse Point Cloud Completion and Surface Reconstruction of High-Speed Train Noses

Project Lead, Supervised by Prof. Zhao Tang, State Key Laboratory of Rail Transit Vehicle System Apr 2024 - Mar 2025

- Built a portable point cloud acquisition system integrating LiDAR, a camera, and an IMU, and deployed FAST-LIVO2 for sensor fusion, pose estimation, and mapping.
- Established a point-cloud processing pipeline using moving least squares for smoothing and geometric completion to address missing observations in glass-window regions.
- Proposed a completion method combining tri-axial projection, section-based interpolation, and reverse hole-boundary identification to recover missing geometry from surrounding observations.
- Compared greedy projection triangulation, Poisson reconstruction, and Marching Cubes-based reconstruction, examining window-region recovery, surface continuity, and overall shape preservation.
- Research output: Preprint posted on arXiv (arXiv:2610.05758).

PREPRINT

Ma, X., Tang, Z., Lai, H., Chen, R., & Jin, J. (2026). A Three-Dimensional Reverse-Projection Method for Sparse Point Cloud Completion and Its Application to High-Speed Train Nose Reconstruction. *arXiv Preprint*. <https://doi.org/10.48550/arXiv.2610.05758>

PROJECT EXPERIENCES

FPGA-Based Digital System Design May 2026 - Jun 2026

- Traffic-light controller development in Quartus/HDL, logic simulation, waveform analysis, and hardware testing.

STM32-Based Embedded Game Development Apr 2026 - Jun 2026

- Embedded game development using STM32 hardware and peripherals.

Audio-Visualisation PCB Design and Fabrication Mar 2026 - May 2026

- Three-band LED schematic and PCB layout in KiCad, component soldering, and multimeter testing.

Emerging Optoelectronic System Design Oct 2025 - Mar 2026

- Joint optoelectronic system design with University of Leeds students.

Line-Following Model Car & Washing Machine Controller Apr 2025 - May 2025

- STM32 programming in Keil5, MOSFET motor control, and car-body assembly

Washing Machine Controller Apr 2025 - May 2025

- STM32 programming in Mbed, mode and timer control, child lock, and sensor responses.

HONORS AND AWARDS

- Second Prize (AI Innovation Track), 27th China Robot & Artificial Intelligence Competition - National Finals, 2025. (An Online Sports Course Evaluation System Based on Multi-Expert Collaborative Systems and Sentiment Analysis)
- Outstanding Student Leader, SWJTU-Leeds Joint School, Southwest Jiaotong University, 2025.
- Mingcheng Award, SWJTU-Leeds Joint School, Southwest Jiaotong University, 2024.

EXTRACURRICULAR ACTIVITIES AND VOLUNTEER

Academic Innovation Base of SWJTU-Leeds Joint School, Southwest Jiaotong University Sep 2023 - May 2026

Roles: Member (2023-2024); Head of Activities (2024-2025); President (2025-2026)

- As president, coordinated over 30 members and 4 departments, liaised with more than 80 guests, planned event schedules, venues, and themes, and spearheaded the development of the “Leeds Tech Month” flagship competition, attracting over 500 participating teams, with events reaching more than 3,000 faculty and students.

TECHNICAL SKILLS AND ADDITIONAL INFORMATION

Technical Skills: C/C++/Python/MATLAB(VS2017, MbedStudio, OnlineGDB, Keil5); ROS2; Linux; SLAM (FAST-LIO/FAST-LIVO2/R3LIVE); point-cloud/3D reconstruction; VTK/PCL/OpenCV; embedded integration (STM32).

Languages: Mandarin(Native), English(Fluent). **Interests:** Eight Brocades, Simplified 24-Form Tai Chi Chuan